

Variation in *E. coli* Measurements at Toronto Beaches*

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This paper examines variation in *E. coli* measurements at Toronto beaches using 102,534 observations collected from 2007 to 2026. The measurements are strongly right-skewed, with most observations concentrated at relatively low values and a small number of extremely high values. *E. coli* measurements also differ across beaches, with Sunnyside Beach and Marie Curtis Park East Beach generally having higher values than several other beaches. Measurements vary over time, but there is no clear sustained upward or downward pattern over the study period.

1 Introduction

Toronto’s public beaches provide important recreational spaces during the summer swimming season, but water quality can vary across locations and over time. One commonly used indicator of recreational water quality is the concentration of *Escherichia coli* (*E. coli*), which is monitored by Toronto Public Health to assess conditions at supervised beaches (City of Toronto, n.d.). *E. coli* is also used as an indicator of fecal contamination in recreational fresh water (Health Canada 2023). Understanding how *E. coli* measurements are distributed and how they differ across beaches and over time can provide insight into variation in recreational water quality across Toronto.

Using Toronto Beaches Water Quality data from 2007 to 2026, this paper examines the distribution of *E. coli* measurements across Toronto beaches. The measurements are strongly right-skewed: most observations are concentrated at relatively low values, while a small number are extremely high. Measurements also differ across beaches, with Sunnyside Beach and Marie Curtis Park East Beach generally having higher values than several other beaches.

*Code and data are available at: <https://github.com/Sophiaaa12/STA2453-assignment-1>.

These patterns matter because *E. coli* is used to assess recreational water quality. Recent Canadian research has also examined the relationship between *E. coli* levels and acute gastrointestinal illness among beachgoers (Young et al. 2026). Examining the overall distribution of measurements and differences across beaches and over time provides a more complete description of the water-quality data and highlights the substantial variation present in the observations.

The remainder of this paper is structured as follows. Section 2 describes the Toronto Beaches Water Quality dataset, explains how the *E. coli* measurements are collected and recorded, and examines their overall distribution and variation across beaches and over time.

2 Data

2.1 Overview

The Toronto Beaches Water Quality dataset is provided through the City of Toronto Open Data Portal (City of Toronto, n.d.). The data were downloaded reproducibly using the `opendatatoronto` R package (Gelfand 2025). During the summer swimming season, the City collects water samples from supervised public beaches, and Toronto Public Health analyzes the samples for *E. coli* as part of its monitoring of beach water quality (City of Toronto, n.d.). These measurements provide information used to assess recreational water conditions at Toronto beaches.

The data used in this analysis contain 102,534 observations collected between June 3, 2007 and September 16, 2026 across 10 Toronto beaches. Each observation records a beach identifier, beach name, sampling site, collection date, and *E. coli* measurement. The original dataset also contains an observation identifier and geographic information, which are not required for this analysis and were removed during data cleaning. The analysis was conducted using R (R Core Team 2024), the `tidyverse` (Wickham et al. 2019), and the `here` package (Müller 2020). Figures were produced using `ggplot2` (Wickham 2016), and tables were produced using `tinytable` (Arel-Bundock 2026).

2.2 Measurement

Each row in the analysis data represents an *E. coli* measurement from a particular sampling site at a Toronto beach on a particular collection date. Multiple observations can be recorded for the same beach and date because samples are collected at different sites. Therefore, an observation should not be interpreted as the overall water quality of an entire beach on that day. *E. coli* is used by Toronto Public Health as an indicator when monitoring recreational water quality (City of Toronto, n.d.).

Table 1: Summary of E. coli measurements in the analysis data

Statistic	Value
Number of observations	102,534
Number of beaches	10
Median E. coli	20
Mean E. coli	154
Minimum E. coli	1
Maximum E. coli	6,191,768
Missing E. coli	3,543

Table 1 summarizes the distribution and the number of missings of E. coli measurements in the analysis data. The measurements are strongly right-skewed, with a median of 20 and a mean of approximately 154. The largest recorded value is 6,191,768, illustrating the presence of extreme observations. Because the mean is sensitive to these unusually large values, the median provides a more representative description of a typical observation. In addition, 3,543 E. coli measurements are missing. These observations are retained in the cleaned dataset and excluded only from calculations or visualizations that require an observed E. coli value.

There are limited direct privacy concerns because the dataset contains environmental measurements rather than information about individuals. However, the measurements have public-health relevance and should therefore be interpreted carefully. Individual site-level observations are not classified as “safe” or “unsafe” in this analysis because they do not necessarily represent the water quality of an entire beach.

2.3 Data exploration

2.3.1 Overall distribution

The overall distribution of E. coli measurements is strongly right-skewed, as shown in Figure 1. Most observations are concentrated at relatively low values, while progressively fewer observations occur at higher values. A logarithmic scale is used on the horizontal axis so that the large range of observed values can be displayed more clearly.

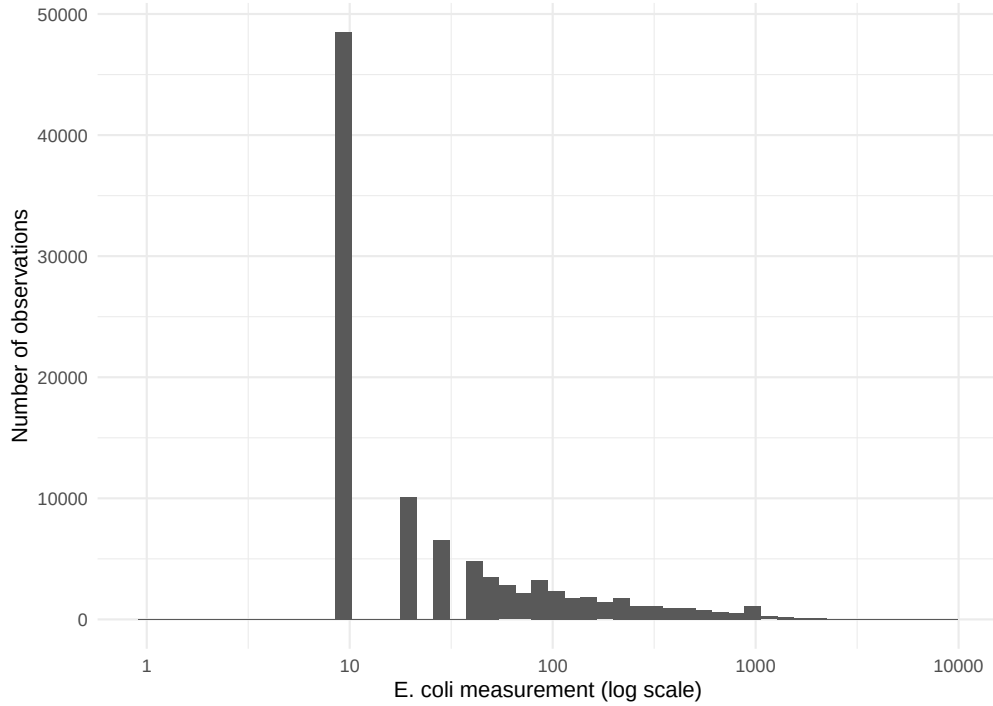


Figure 1: Overall distribution of E. coli measurements up to 10,000

Figure 1 displays measurements up to 10,000 so that the distribution of the large majority of observations remains visible. Measurements above this value are retained in the analysis data and are included in the numerical summaries in Table 1. The concentration of observations at lower values together with a long upper tail is consistent with the large difference between the median and mean reported in Table 1. This skewness also means that a small number of unusually high measurements can have a substantial influence on the mean, making the median more informative for describing a typical observation.

2.3.2 Variation across beaches

E. coli measurements differ across Toronto beaches, as shown in Figure 2. Each point represents an individual E. coli measurement, and a logarithmic scale is used to display the wide range of observed values more clearly.

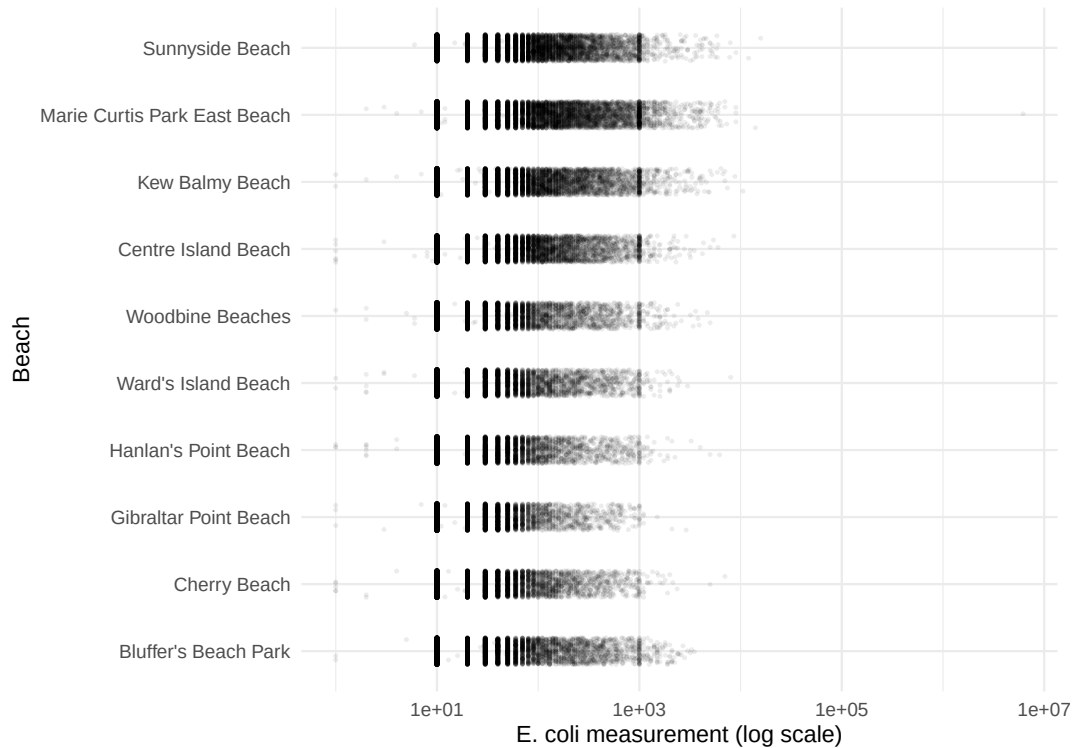


Figure 2: E. coli measurements across Toronto beaches

Figure 2 shows substantial variation both across beaches and among observations from the same beach. Sunnyside Beach and Marie Curtis Park East Beach have more observations extending into higher E. coli values, while several other beaches have measurements more concentrated at lower values. However, the measurements overlap considerably across beaches, and relatively high values can be observed at all beaches. This suggests that differences across beaches are present, but there is also substantial variation within individual beaches.

2.3.3 Variation over time

E. coli measurements also vary over time, as shown in Figure 3. Each point represents an individual E. coli measurement collected between 2007 and 2026. A logarithmic scale is again used to display the wide range of observed values more clearly.

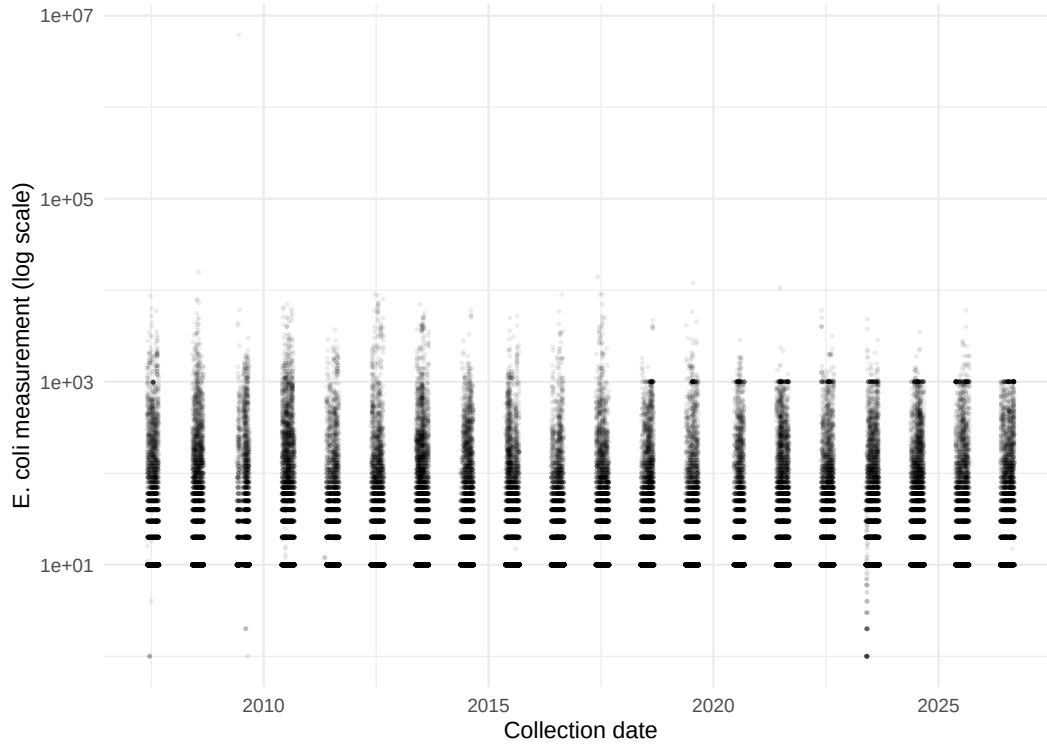


Figure 3: E. coli measurements over time

Figure 3 shows substantial variation in E. coli measurements throughout the study period. Most observations are concentrated at relatively low values, while higher measurements occur in each year. There is an extreme observation point appeared around 2009, almost reaching 10^7 . Overall, there is no clear sustained upward or downward pattern in the observed measurements over time.

References

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